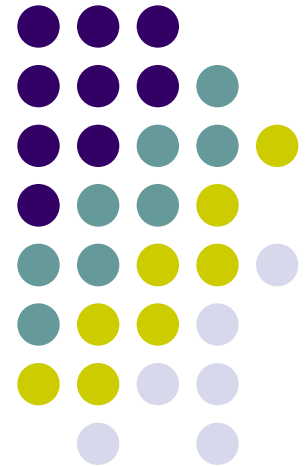
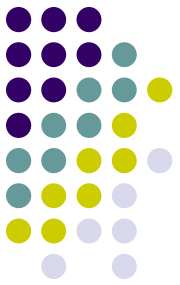


Database Systems and Data-Driven Applications

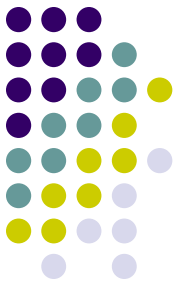
ORDB: Triggers and ER-ORDB
Mapping





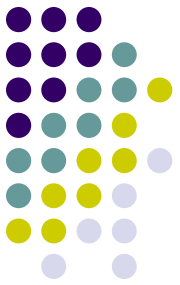
Last Week

- Methods
 - Member methods
 - Comparison methods (MAP & ORDER)
 - Methods on nested tables
- Inheritance
 - Type inheritance (UNDER, FINAL, NOT FINAL)
 - INSTANTIABLE/NOT INSTANTIABLE Types and Methods
 - Methods (Overriding, Overloading)
- Any questions?



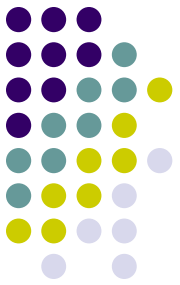
Triggers

- Procedures stored in the database and implicitly run, or *fired*, when something happens.
 - INSERT, UPDATE, or DELETE on a table or view (DML Triggers).
 - System and other data events on DATABASE and SCHEMA (System Triggers).



Triggers on Object Tables

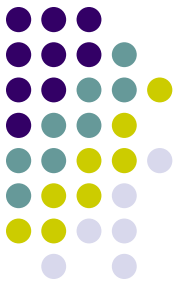
- Triggers can be defined on an object table just as on other tables.
- Example:
 - CREATE TYPE location as OBJECT (campus VARCHAR2(20), building NUMBER, room NUMBER);
 - CREATE TYPE staff as OBJECT(sid NUMBER, name VARCHAR2(100), address VARCHAR2(100), office location);
 - CREATE TABLE staff_tab OF staff (sid PRIMARY KEY);



Example

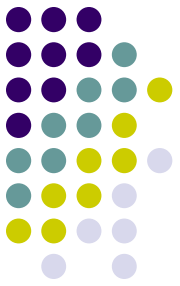
- CREATE TABLE movement (sid NUMBER, old_office location, new_office location);
- CREATE TRIGGER trig1
BEFORE UPDATE OF office ON staff_tab
FOR EACH ROW
WHEN (new.office.campus = 'Bentley')
BEGIN
 IF :new.office.building = 314 THEN
 INSERT INTO movement VALUES (:old.sid,:old.office,
:new.office);
 END IF;
END;

Database design for ORDB



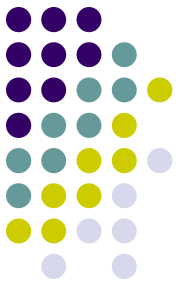
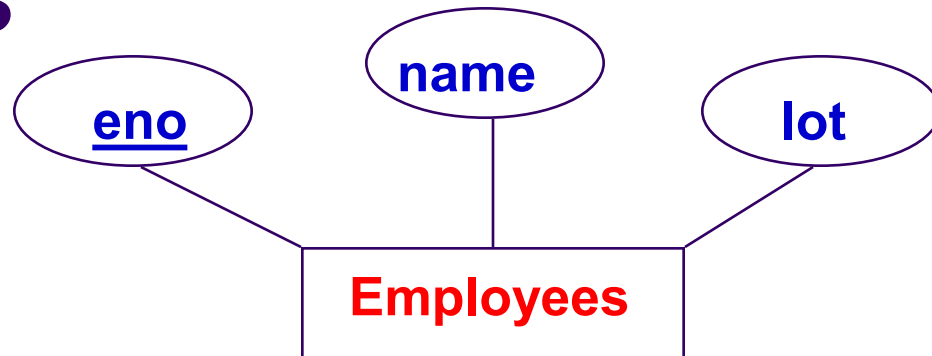
- ER model may be extended with more constructs such as:
 - Multi-valued attributes (e.g., locations of a store)
 - Structured-type attributes (e.g., address made up of house no, street, suburb, etc.)
 - Collection-type attributes (e.g., list of dates)

Mapping ERD to ORDB



- Map each ER entity type to ORDB type
 - Include all attributes of ER type in ORDB type
 - Declare a table for each ORDB type and specify key attributes as (primary) keys
 - Sub-types in ISA hierarchy inherit attributes of super type

Entity Types

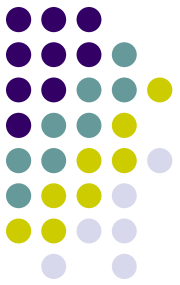
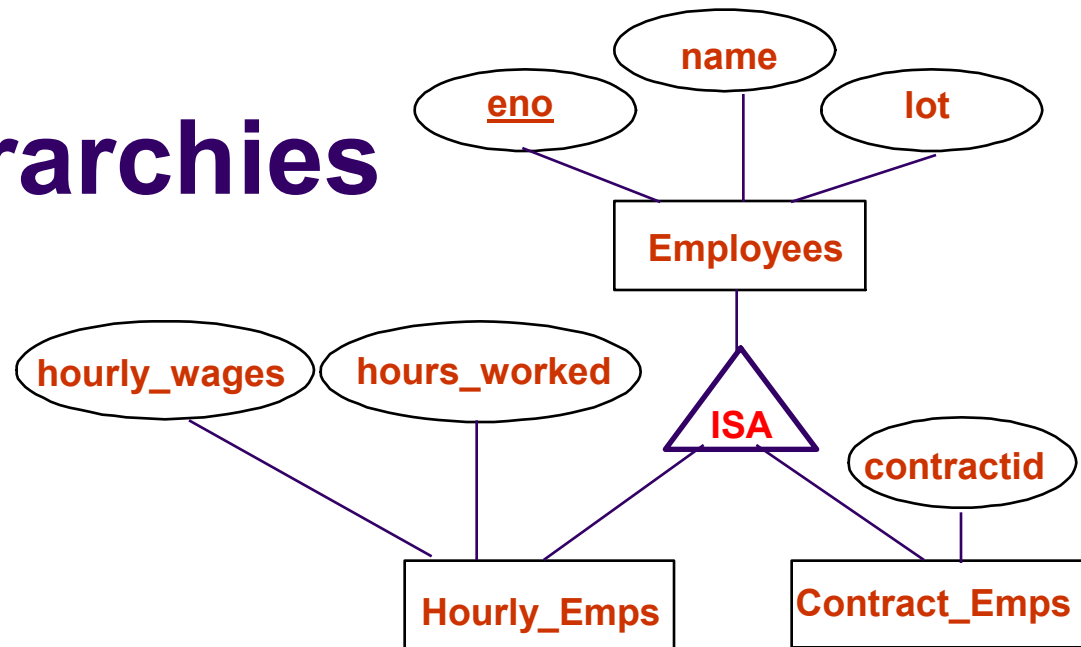


- Entity type can be mapped to a type easily.

```
CREATE TYPE Emp_t  
    (eno CHAR(11),  
     name CHAR(20),  
     lot INTEGER);
```

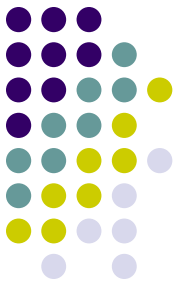
```
Create table employees of Emp_t(  
    eno primary key);
```


ISA Hierarchies

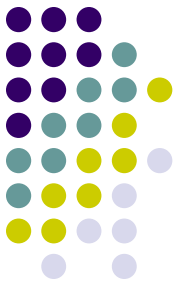


```
CREATE TYPE emp_t AS OBJECT (...) NOT FINAL;  
CREATE TYPE hourly_ems_t UNDER emp_t  
    (hourly_wages number(6,2),  
     hours_wkd number(4,2));  
CREATE TYPE contract_ems_t UNDER emp_t  
    (contractid char(10));
```

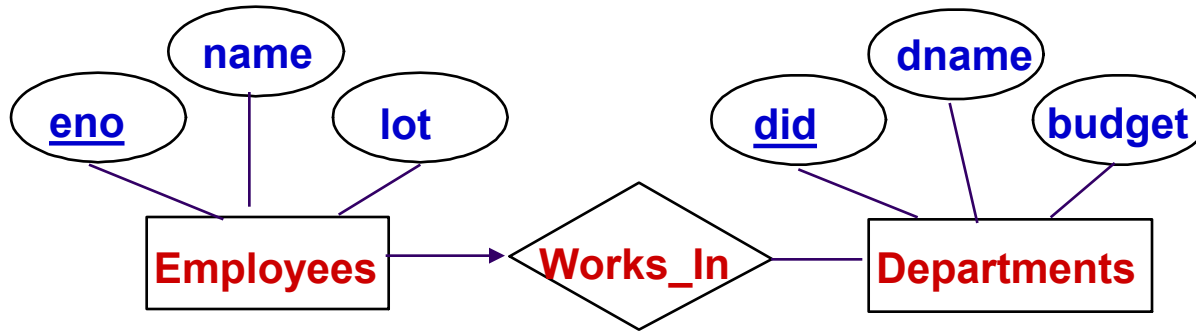
Mapping ERD to ORDB



- Binary relationship types
 - Cardinality 1:1, N:1, or M:N
- Weak entity types
 - Similar to regular entity types
 - Alternative mapping possible if the weak entity does not participate in any other relationship
- N-ary relationships with $n > 2$
 - Mapped as a separate type
 - Appropriate references to each participating class
- Methods for types (not in ER)
 - UML?

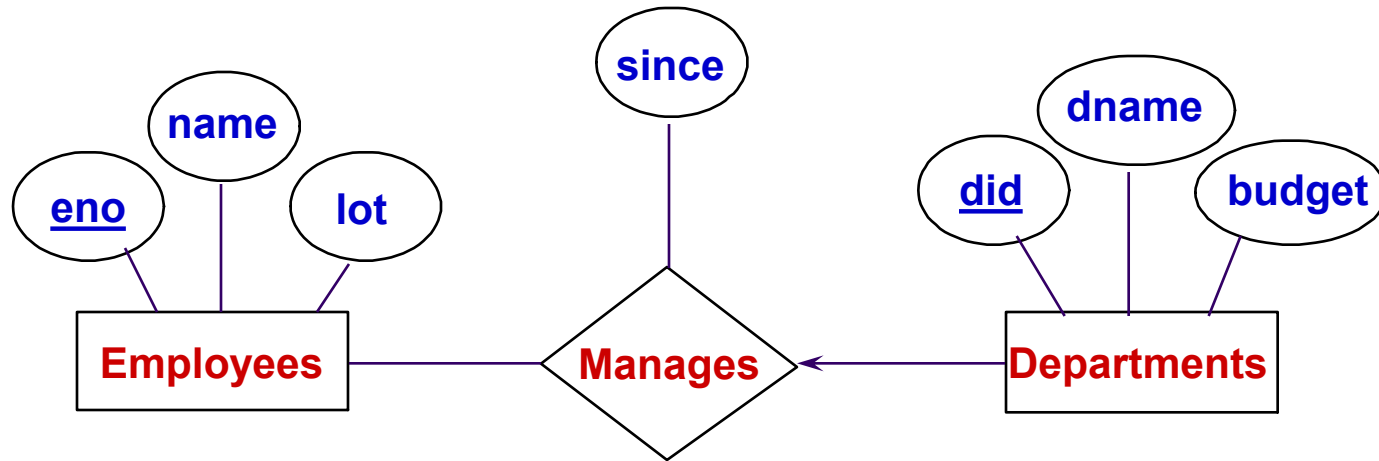
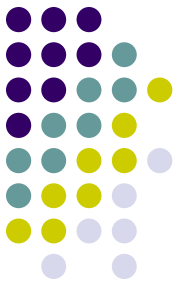


Relationship sets

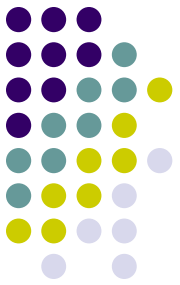


- Add a reference attribute for a binary relationship with key constraint into the corresponding object type.
 - `CREATE TYPE Emp_t`
 `(eno CHAR(11),`
 `name CHAR(20),`
 `lot INTEGER,`
 `workdept ref dept_t);`

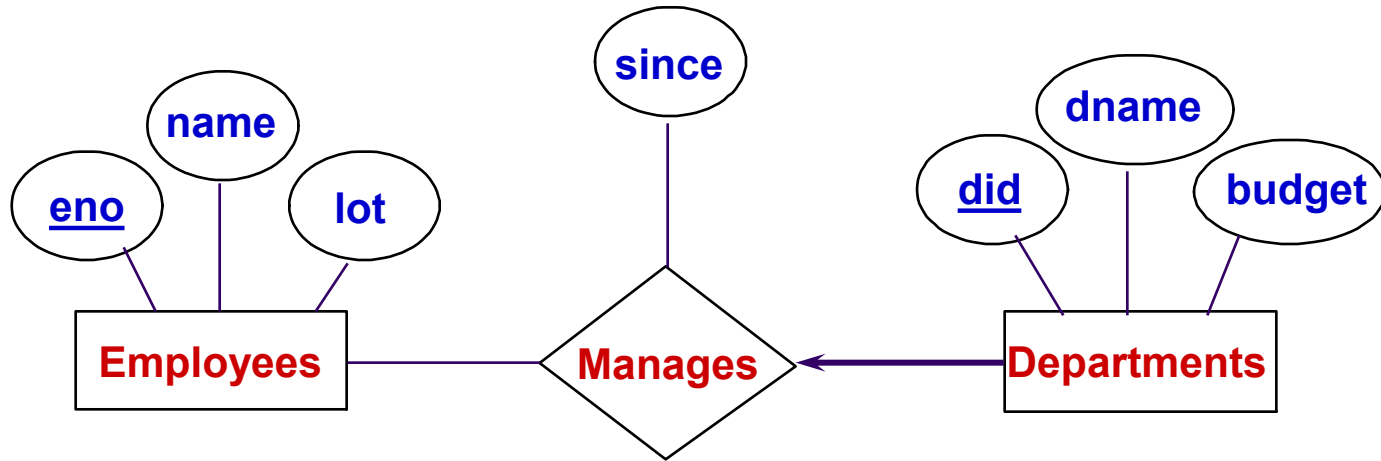
Translating Key Constraints



```
CREATE TYPE Dept_t as object
( did INTEGER,
  dname CHAR(20),
  budget REAL,
  mgr ref emp_t
  since DATE);
```

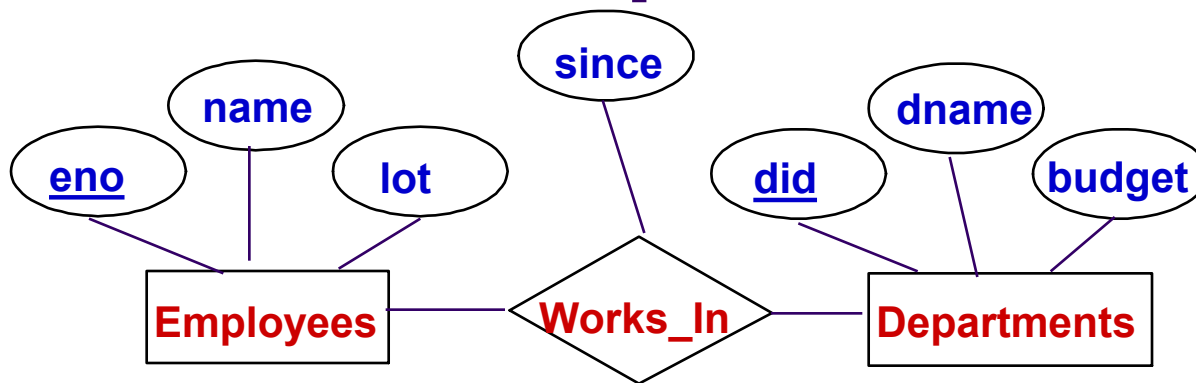


Participation Constraints



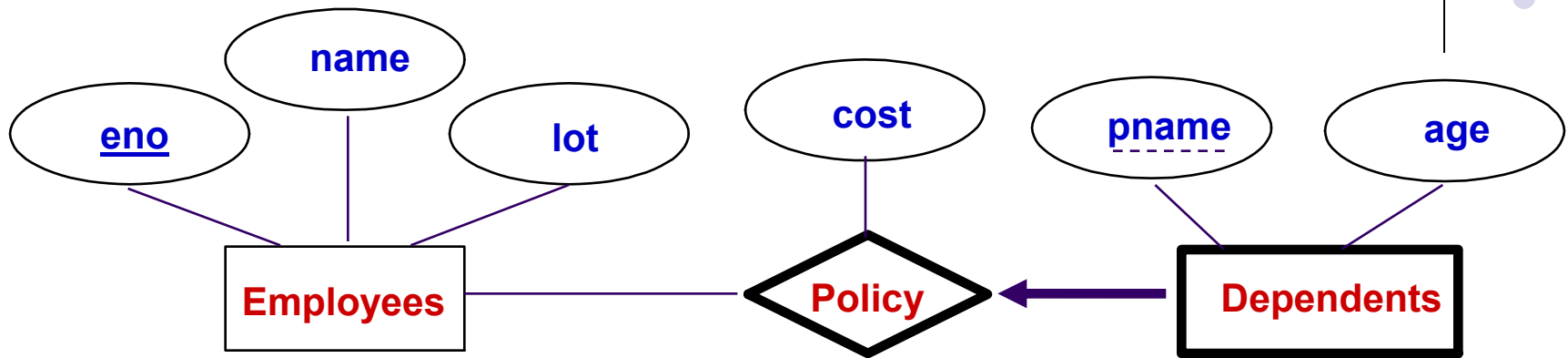
```
CREATE TABLE Department of Dept_t  
( did primary key,  
  mgr NOT NULL REFERENCES Employees  
);
```

M:N Relationship set



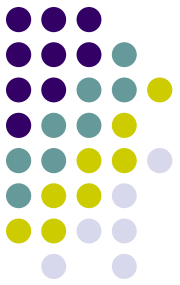
```
CREATE TYPE WorksIn_t AS OBJECT(  
    emp REF emp_t,  
    dept REF dept_t,  
    since DATE);  
CREATE TABLE worksIn OF WorksIn_t(  
    emp REFERENCES Employees,  
    dept REFERENCES Departments);
```

Weak Entities



- Option 1: Map like a regular entity
- Option 2: Weak entity types that do not participate in any relationships except their identifying relationship with owning entity.
 - Use nested collection type for the weak entity information.
- Option 3: Weak entity type objects are not to be referenced by attributes of any other object.
 - Use nested collection type for the weak entity information.

Weak Entity Sets: Option 2



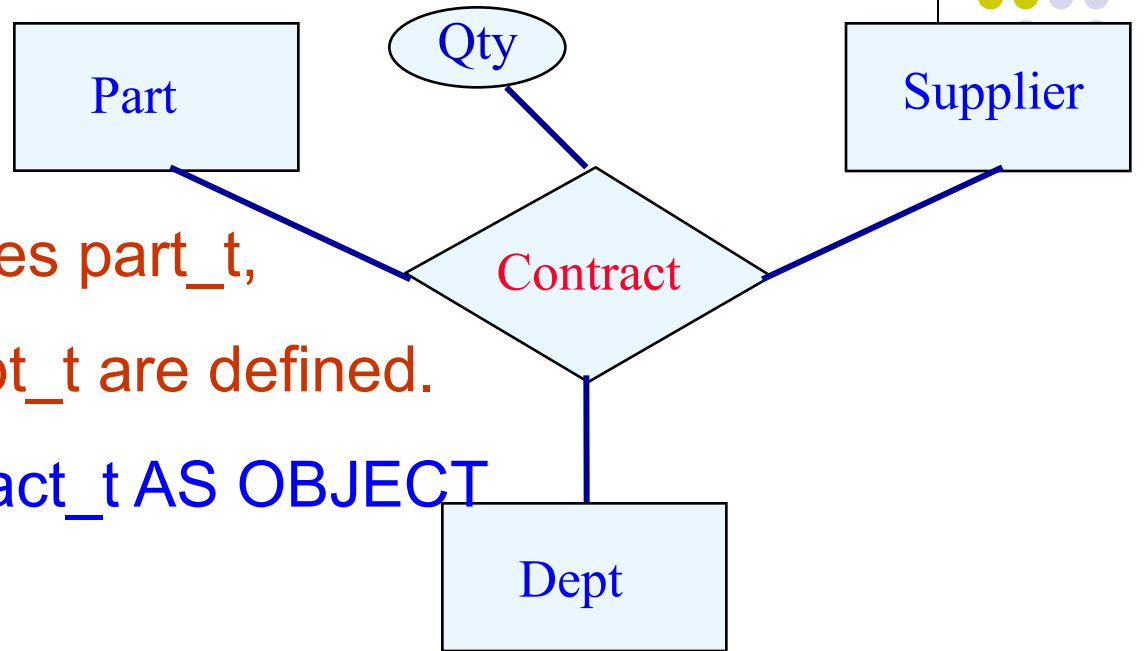
Create type depend_t as object(
 Pname varchar(12),
 Age integer,
 policyCost number (6,2));

Create type policy_t as table of depend_t;

Create type emp_t as object(...,

 dependents policy_t)

N-ary relationship



➤ Assuming object types `part_t`,
`supplier_t`, and `dept_t` are defined.

```
CREATE TYPE Contract_t AS OBJECT
```

```
(part REF part_t,
```

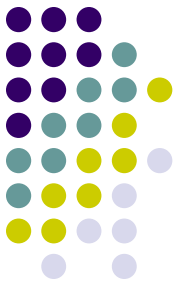
```
supplier REF supplier_t,
```

```
dept REF dept_t,
```

```
qty NUMBER(6,2)
```

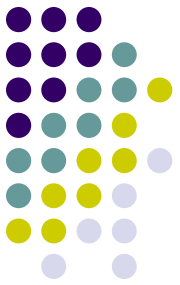
```
);
```





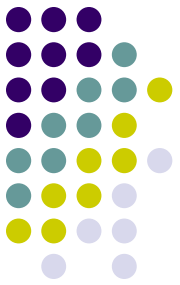
Mapping to OR Schema

- Regular entities to types
 - Customer_typ (custNo, ...)
 - StockItem_typ (StockNo, ...)
 - PurchaseOrder_typ (PONo, ...)
- Inheritance
 - StockItem_typ (StockNo, ...) NOT FINAL
 - Book_typ(...) under StockItem_typ
 - Software_typ(...) under StockItem_typ



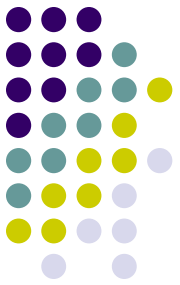
Mapping to OR Schema

- Weak entity type
 - Lineltem_typ (LineltemNo, ...)
 - LineltemList_ntabtyp AS TABLE OF Lineltem_typ
 - PurchaseOrder_typ (PONo, ..., LineltemList_ntab LineltemList_ntabtyp)
- Binary relationship with key constraint
 - PurchaseOrder_typ (PONo, ..., LineltemList_ntab LineltemList_ntabtyp, Cust_ref REF Customer_typ)
 - Lineltem_typ (LineltemNo, ..., Stock_ref REF StockItem_typ)



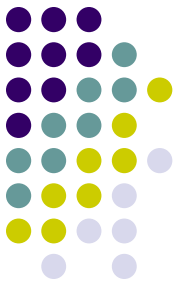
Final types

- Customer_typ (custNo, ...)
- StockItem_typ (StockNo, ...)
- Book_typ(...) under StockItem_typ
- Software_typ(...) StockItem_typ
- Lineltem_typ (LineltemNo, ...,
Stock_ref REF StockItem_typ)
- LineltemList_ntabtyp AS TABLE OF Lineltem_typ
- PurchaseOrder_typ (PONo, ...,
LineltemList_ntab LineltemList_ntabtyp,
Cust_ref REF Customer_typ)



Tables and constraints

- Customers of Customer_typ (custNo primary key)
- Stocks of StockItem_typ (StockNo primary key)
- Orders of PurchaseOrder_typ (PONo primary key, Cust_ref references Customers)
nested table LineItemlist



Summary

- Trigger Example
- Mapping ER to ORDB
 - Mapping various ER features to ORDB
 - Example